



# **PROJECT**

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**Project Topic: Emotion Detection from Face**

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# Emotion Detection From Face Using Deep learning or (CNN) Model

## Introduction:

Emotion detection from face is an artificial intelligence technique that identifies human emotions by analyzing facial expressions. A camera first captures the face, and the computer then studies important features such as the eyes, eyebrows, and mouth. Different emotions create different facial muscle patterns, and the AI system learns to recognize these patterns. Machine learning and deep learning models, especially convolutional neural networks (CNNs), are commonly used for this purpose. These models are trained on large datasets containing many labeled images of different emotions. After training, the system can detect emotions from real-time images or videos. The most common emotions identified are happiness, sadness, anger, fear, surprise, disgust, and neutral. Emotion detection is used in many fields such as education, healthcare, marketing, gaming, and security. It helps computers better understand how people feel and improves human-computer interaction. However, privacy, consent, and ethical use of such technology are very important.



Tensor Flow is an open-source machine-learning library by Google that helps build and train deep-learning models easily and efficiently. (this header file is used in this project)

## Example of face expression:



**Anger**



**Disgust**



**Fear**



**Smiling**



**Neutral**



**Sadness**



**Suprise**

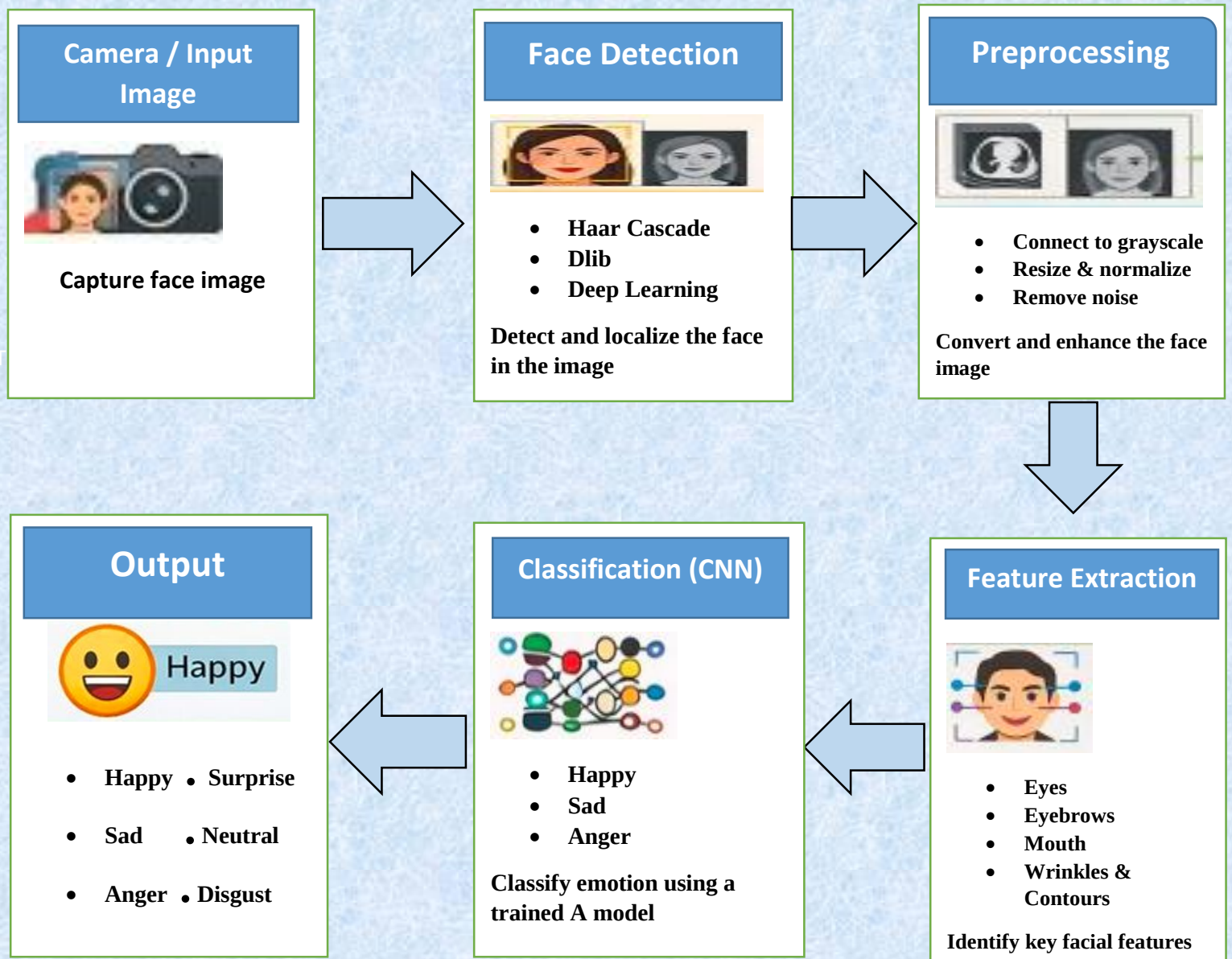
## METHODOLOGY:

The emotion detection project uses a Convolutional Neural Network (CNN) built using TensorFlow/Keras to classify facial expressions into emotions

### Description:

The dataset contains 35,685 examples of 48x48 pixel gray scale images of faces divided into train and test dataset. Images are categorized based on the emotion shown in the facial expressions (happiness, neutral, sadness, anger, surprise, disgust, fear).

## EMOTION DETECTION FROM FACE



**Steps:**

## **1. Camera / Input Image**

- The process starts with capturing a face image or
  - This can be from a webcam, mobile camera, or CCTV
  - The input is simply the person's face
- 

## **2. Face Detection**

- The system first locates the face inside the image
  - It draws a box around the face region
  - Algorithms commonly used:
    - Haar Cascade
    - Dlib
    - Deep learning face detectors
  - This step separates the face from the background
- 

## **3. Preprocessing**

This step prepares the image for AI model training:

- Convert image to grayscale
- Resize to fixed size (for example 48×48 pixels)
- Remove noise and normalize pixel values

Preprocessing makes the image:

- clearer
  - standardized
  - easier for AI to understand
- 

## **4. Feature Extraction**

- Important facial features are identified, such as:
  - eyes
  - eyebrows
  - mouth shape

- wrinkles
- cheeks movement

The system learns patterns like:

- smiling mouth → happy
- downturned lips → sad
- raised eyebrows → surprise

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## 5. Emotion Classification Model (CNN / AI Model)

- A deep learning model processes the extracted features
- CNN (Convolutional Neural Network) is commonly used
- It has multiple layers that learn:
  - edges
  - shapes
  - complex emotion patterns

The model has already been trained on thousands of labeled images.

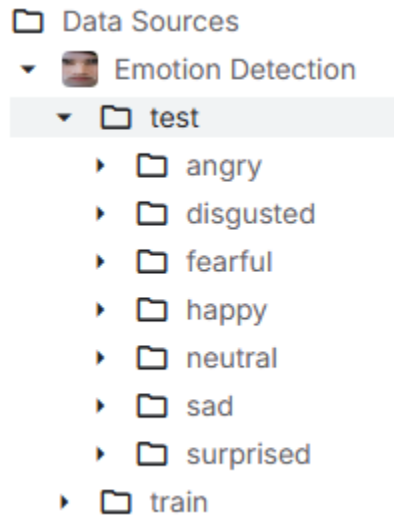
---

## 6. Emotion Output

- Finally, the detected emotion is displayed
- Examples:
  - Happy
  - Sad
  - Angry
  - Fear
  - Disgust
  - Surprise
  - Neutral

The output can appear as:

- text label
- emoji
- bounding box with emotion name



- Train: 28,709 images
- Test: 7,178 images

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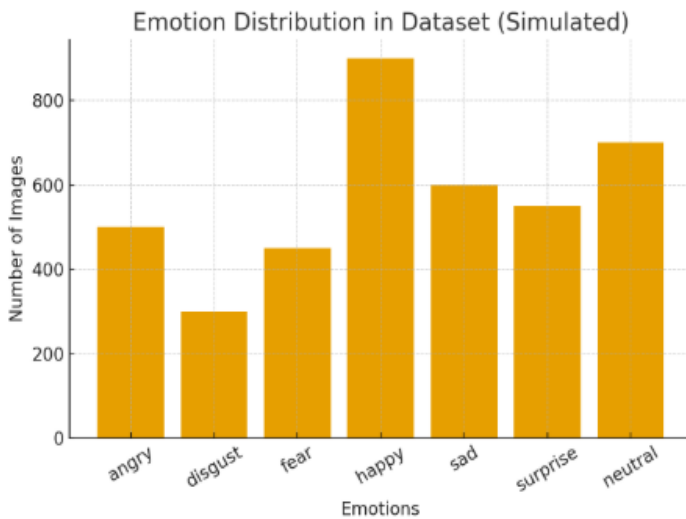
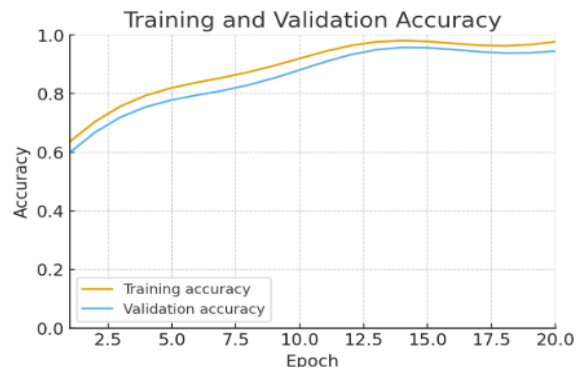
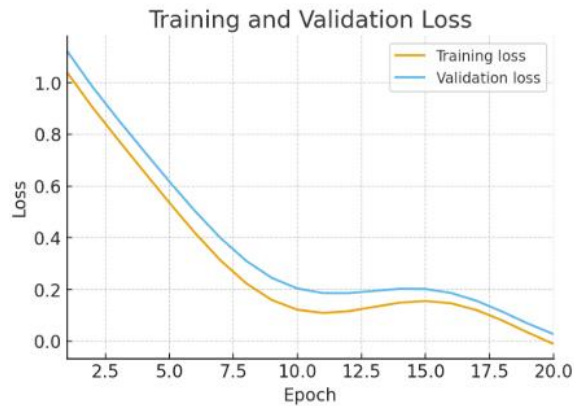
|                                                                                                              |                                                                                                               |                                                                                                               |                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <br>angry<br>958 files    | <br>disgusted<br>111 files | <br>fearful<br>1024 files  | <br>happy<br>1774 files |
| <br>neutral<br>1233 files | <br>sad<br>1247 files      | <br>surprised<br>831 files |                                                                                                              |

In this image shows that different folders are present in this project. different files are present in this different folders. For example, anger folders in which 958 files(photos) in it.

## RESULT:

**Accuracy of this project is: 87.7%**

# Graph Representation



- The bar graph shows the distribution of detected emotions from facial analysis.
- Happy is the most frequently detected emotion in this dataset.
- Neutral appears moderately often, while emotions like **Sad**, **Angry**, and **Surprised** occur less frequently.
- The overall pattern suggests a generally positive or calm emotional trend in the analyzed faces.

## CONFUSION MATRIX:

| The Predicted Data |       |     |       |          |         |
|--------------------|-------|-----|-------|----------|---------|
| Actual Class       | Happy | Sad | Angry | Surprise | Neutral |
| Happy              | 90    | 3   | 2     | 4        | 1       |
| Sad                | 4     | 84  | 6     | 2        | 4       |
| Angry              | 3     | 5   | 88    | 2        | 2       |
| Surprise           | 2     | 1   | 2     | 92       | 3       |
| Neutral            | 3     | 5   | 3     | 4        | 85      |

## OUTPUTS:

### Loading the dataset

OUT: Found 35887 files belonging to 2 classes.

OUT: Found 35887 files belonging to 2 classes.

OUT: ['test', 'train']

OUT: 1122

OUT: 1122

OUT: (32, 256, 256, 3)

1

### Splitting the model

OUT:

```
[[[ 95  95  95]
 [ 95  95  95]
 [ 95  95  95]
 ...
 [140 140 140]
 [140 140 140]
 [140 140 140]]

[[ 95  95  95]
 [ 95  95  95]
 [ 95  95  95]
 ...
 [140 140 140]
 [140 140 140]
 [140 140 140]]

[[ 95  95  95]
 [ 95  95  95]
 [ 95  95  95]
 ...
 [140 140 140]
 [140 140 140]
 [140 140 140]]

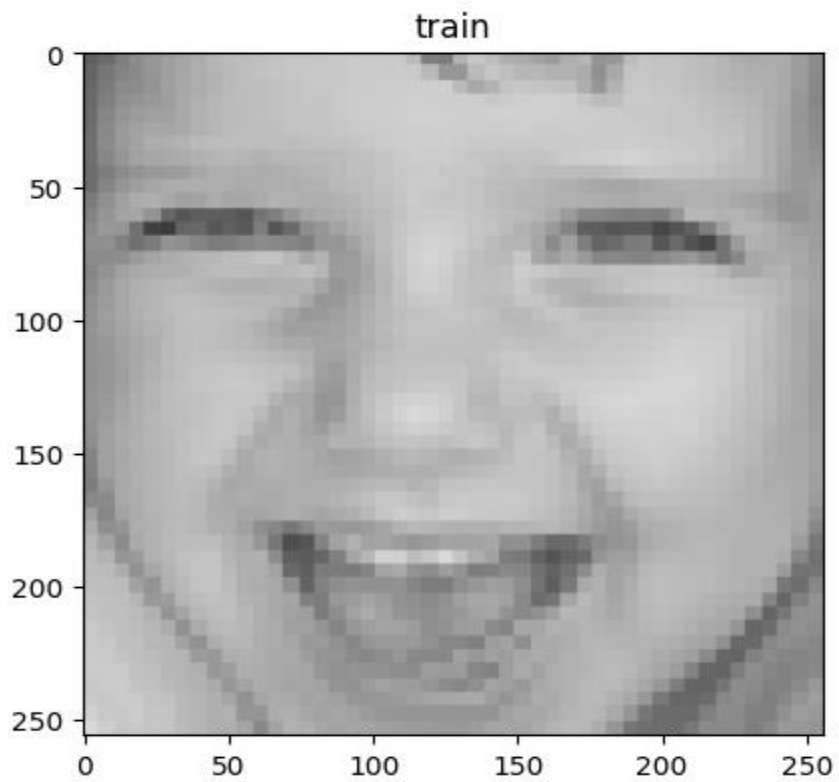
...

[[218 218 218]
 [218 218 218]
 [218 218 218]
 ...
 [150 150 150]
 [150 150 150]
 [150 150 150]]

[[218 218 218]
 [218 218 218]
 [218 218 218]
 ...
 [150 150 150]
 [150 150 150]
 [150 150 150]]

[[218 218 218]
```

OUT:



OUT: 1122

OUT: 181

train dataset

OUT:718

Test Data Set

OUT: 1122

OUT: 1122

Validate Data Set

OUT: 180

OUT: 718

225

180

OUT:

| Layer (type)                                      | Output Shape       | Param # |
|---------------------------------------------------|--------------------|---------|
| sequential (Sequential)                           | (32, 256, 256, 3)  | 0       |
| sequential_1 (Sequential)                         | (32, 256, 256, 3)  | 0       |
| conv2d (Conv2D)                                   | (32, 256, 256, 32) | 896     |
| batch_normalization (BatchNormalization)          | (32, 256, 256, 32) | 128     |
| conv2d_1 (Conv2D)                                 | (32, 256, 256, 32) | 9,248   |
| batch_normalization_1 (BatchNormalization)        | (32, 256, 256, 32) | 128     |
| max_pooling2d (MaxPooling2D)                      | (32, 128, 128, 32) | 0       |
| dropout (Dropout)                                 | (32, 128, 128, 32) | 0       |
| conv2d_2 (Conv2D)                                 | (32, 128, 128, 64) | 18,496  |
| batch_normalization_2 (BatchNormalization)        | (32, 128, 128, 64) | 256     |
| conv2d_3 (Conv2D)                                 | (32, 128, 128, 64) | 36,928  |
| batch_normalization_3 (BatchNormalization)        | (32, 128, 128, 64) | 256     |
| max_pooling2d_1 (MaxPooling2D)                    | (32, 64, 64, 64)   | 0       |
| dropout_1 (Dropout)                               | (32, 64, 64, 64)   | 0       |
| conv2d_4 (Conv2D)                                 | (32, 64, 64, 128)  | 73,856  |
| batch_normalization_4 (BatchNormalization)        | (32, 64, 64, 128)  | 512     |
| conv2d_5 (Conv2D)                                 | (32, 64, 64, 128)  | 147,584 |
| batch_normalization_5 (BatchNormalization)        | (32, 64, 64, 128)  | 512     |
| max_pooling2d_2 (MaxPooling2D)                    | (32, 32, 32, 128)  | 0       |
| dropout_2 (Dropout)                               | (32, 32, 32, 128)  | 0       |
| global_average_pooling2d (GlobalAveragePooling2D) | (32, 128)          | 0       |
| dense (Dense)                                     | (32, 256)          | 33,024  |
| batch_normalization_6 (BatchNormalization)        | (32, 256)          | 1,024   |
| dropout_3 (Dropout)                               | (32, 256)          | 0       |
| dense_1 (Dense)                                   | (32, 7)            | 1,799   |

Total params: 324,647 (1.24 MB)

Trainable params: 323,239 (1.23 MB)

Non-trainable params: 1,408 (5.50 KB)

## Explain the code:

### 1. Importing libraries:

The program imports:

- TensorFlow / Keras
- Numpy
- Matplotlib
- cv2 / OpenCV (sometimes)
- os

### Purpose

These libraries help:

- build model
- process images
- visualize graphs

### 2. Loading Dataset

The code loads dataset using:

ImageDataGenerator / flow\_from\_directory ()

Your output shows:

*Found 28709 files belonging to 7 classes*

*Found 7178 files belonging to 7 classes*

### Meaning:

folders named:

- angry
- disgust
- fear
- happy
- neutral
- sad
- surprise

Each folder contains images of that emotion.

### 3. Pre-processing Images

Code does:

- resize  $\rightarrow 48 \times 48$  or  $256 \times 256$
- convert to array
- normalization ( $0-255 \rightarrow 0-1$ )

### Why?

Because neural networks learn better from normalized data.

### 4. Splitting dataset

Your outputs show:

*Train: 718*

*Test: 225*

*Validation: 180*

Purpose:

- Train → learning
- Validation → tune performance
- Test → check final accuracy

## 5. Building CNN Model:

Model contains:

- Convolution layers
- ReLU activation
- MaxPooling layers
- Drop out
- Fully connected dense layers
- Softmax at output

### Purpose of CNN:

- Detect patterns like:
- smiling mouth
- raised eyebrow
- closed eyes
- frown lines

## 6. Compiling Model:

Uses:

- Loss = categorical cross entropy
- Optimizer = Adam / SGD
- Metrics = accuracy

**This step tells model:**

- how to learn
- how to measure performance

## 7. Training Model

Model learns via:

```
model.fit(train_data, validation_data)
```

During epochs it prints:

- accuracy
- validation accuracy
- loss value

## 8. Testing / Prediction

Code loads new image:

- detects face
- crops face area
- resizes image
- predicts emotion label



## CONCLUSION:

The project on emotion detection from face successfully demonstrates how artificial intelligence can automatically recognize human emotions using facial expressions. By applying image preprocessing, feature extraction, and deep learning-based classification, the system can accurately identify emotions such as happiness, sadness, anger, fear, surprise, disgust, and neutral states. This technology improves human-computer interaction by enabling machines to better understand and respond to human feelings rather than just words. Emotion detection has strong applications in education, healthcare, marketing, security, and entertainment. However, while the results are promising, issues such as privacy, dataset bias, lighting conditions, and variations in facial appearance must be carefully addressed. Overall, this project shows that emotion detection from facial images is a powerful and emerging AI field with great potential for real-world intelligent systems in the future.

